

Research paper

SU-VPDN: A scene understanding method for vehicle part detection

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ABSTRACT

High-accuracy positioning and identification of vehicle parts are crucial for status detection, fault diagnosis, and the development of intelligent damage detection systems. However, detecting left-right and front-rear categories in vehicle parts remains challenging due to numerous categories and feature similarity. To address this, we propose the scene understanding vehicle part detection network (SU-VPDN). It comprises three subnetworks: vehicle orientation scene classification (VOSC), vehicle part instance segmentation (VPIS), and scene understanding (SU). VOSC performs preliminary classification of vehicle orientation scenes, while VPIS detects vehicle parts. SU fine-tunes VOSC results using the part logic correction module, fuses VOSC and VPIS results using the part replacement module, and obtains accurate VPIS results through confidence regulation and hierarchical filtering. SU-VPDN integrates results from multiple subnetworks, transforming the challenge of detecting similar objects into orientation scene detection for precise recognition. We evaluate the proposed method on a dataset containing 59 types of vehicle parts and compare it with advanced segmentation detection models. Experimental results demonstrate that SU-VPDN outperforms other models, achieving a 12.2% and 12.3% improvement in mean average precision (mAP) for box and mask, respectively, compared to the baseline model. Moreover, qualitative and quantitative experiments show that SU-VPDN significantly enhances the detection performance of objects with similar features. Overall, our proposed SU-VPDN offers a promising solution for detecting left-right and front-rear categories in vehicle parts. It surpasses existing models and can enable insurance companies to replace damaged parts and improve claims efficiency efficiently.

1. Introduction

As the number of vehicles on the roads continues to increase, the incidence of vehicle collisions is becoming more common in society today. Traditionally, insurance company staff have conducted vehicle scene investigations, requiring large amounts of human resources and resources to complete (Jin, 2012; Ma and Gao, 2007). Based on the statistics obtained from pertinent insurance companies, it has been identified that a significant portion of claims comprise minor and uncomplicated accidents, constituting approximately 72.2% of the total. Suppose an on-site survey is still necessary to determine the extent of the loss. In that case, the insurance company must allocate a considerable expense, estimated at 150 yuan per survey (Xie, 2019). By adopting an intelligent vehicle damage determination system (Singh et al., 2019), it is now possible to reduce claims costs and simplify the entire process in the insurance industry, thereby advancing vehicle insurance technology. Upon the implementation of

online business processing by insurance companies, the need to dispatch surveyors to accident scenes for handling purposes becomes unnecessary. This progression significantly enhances case management efficiency while concurrently reducing personnel costs. Furthermore, customers are relieved from the inconvenience of awaiting representatives from insurance companies and traffic police at the accident site. Approximately half of the affected parties are compelled to endure wait times exceeding one hour, resulting in both temporal wastage and potential traffic congestion. Conversely, utilizing an online intelligent damage determination system proves advantageous, as it can expedite the process by approximately 30 min compared to traditional survey methods, culminating in mere 5-minute completion periods.

Currently, the existing intelligent vehicle damage determination system in the field of computer vision faces several issues that hinder its effective application in production. These issues include errors in damage and part detection, as well as missed detections. Such limitations prevent the system from being utilized efficiently. Inaccurate

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damage detection, for instance, can result in significant fluctuations in claim amounts. On the other hand, part detection plays a crucial role in determining whether a part needs replacement. Any deviations in this process can obstruct the completion of the claim settlement procedure. Consequently, ensuring accurate and rapid identification and classification of vehicle parts in images is the foundation for an effective intelligent vehicle damage determination system (Yu, 2017; Wu, 2020).

At present, vehicle-related detection is mainly divided into whole vehicle detection (Wang et al., 2023b; Sun et al., 2006; Wang et al., 2023a), aerial vehicle image detection (Weber et al., 2021; Hu, 2021; Lin et al., 2020), and vehicle part detection (Singh et al., 2019). Using whole-vehicle detection techniques for multi-vehicle separation within imagery has been extensively explored. However, integrating aerial vehicle image detection into the intelligent damage determination system poses significant challenges. This paper primarily focuses on investigating and analyzing vehicle part detection technologies applicable to the intelligent damage determination system. Vehicle part detection technology can be categorized into traditional image processing-based algorithms and deep learning-based detection algorithms. Traditional image processing algorithms, such as those described in Chavez-Aragon et al. (2011), Leung and Brian (2004), rely on contour features to extract vehicle parts from images. However, these traditional algorithms have limitations, including poor applicability, low precision, and poor robustness. With the introduction of deep learning methods, researchers have gained significant interest due to their ability to locate and detect vehicle parts with stronger feature extraction and anti-interference capabilities and reduced sensitivity to hyperparameters. Shu et al. proposed a Faster R-CNN vehicle part detection method to constrain the center point of the vehicle part, effectively reducing network complexity, improving training efficiency, and enhancing the detection performance of vehicle parts with low recognition accuracy (Shu, 2017). Xie et al. incorporated attention mechanisms into the convolutional neural network-based image instance segmentation method, addressing the issue of low-quality annotated vehicle datasets and improving vehicle parts' recognition accuracy (Xie, 2019). Yu et al. proposed a deep learning-based vehicle detection and fine-grained classification framework for complex background traffic scene images, which improved the Faster R-CNN framework by modifying the Region Proposal Network (RPN) to propose a more suitable vehicle whole-body detection network model specific to the characteristics of vehicles (Yu, 2017). However, these algorithms primarily use object detection models and instance segmentation models and operate best when there are only a few part categories and types of damage to the vehicle. But as the number of parts and types of damage categories increase and the features between categories become remarkably similar, relying solely on deep learning models may not be sufficient for effective object recognition (Geirhos et al., 2019; Deng et al., 2009).

This paper proposes a novel approach to address the issues of left-right category recognition error, front-back category recognition error, and missed detections and false detection caused by the similarity of features between different categories of vehicle multi-parts. We utilize a deep fusion of deep learning models and orientation scenes to enhance vehicle part detection. We extensively researched the rules for vehicle part scenes based on the shooting perspective in a risk control environment. We introduce the concept of eight-orientation scenes for vehicle parts and use an improved ResNet101 (He et al., 2016) orientation scene classification network to train a dataset for vehicle orientation scenes.

We first fine-tune the detected orientation scene during the image inference process using a part logic correction module. Subsequently, we employ the scene understanding module to fuse the high-quality and accurate vehicle orientation scene into the instance segmentation model of the trained vehicle part detection. We also introduce innovative operations in the scene understanding module, such as part replacement, confidence regulation, and hierarchical filtering. This

approach further improves the classification, localization, and mask matching of vehicle parts, thereby overcoming the challenges of similar features between different categories of vehicle parts.

In summary, this work makes the following three contributions:

1. For the first time, we have identified and detected up to 59 vehicle exterior parts that are relatively new to the research community. Furthermore, we have defined and organized two distinct datasets: the Vehicle Orientation Dataset (VOD), which is specifically employed for image classification purposes; and the Vehicle Multi-Part Dataset (VMPD), designed exclusively for instance segmentation.
2. The integration of vehicle orientation scene recognition with vehicle part detection at a fundamental level is proposed to tackle the problem of category similarity. This approach addresses a critical limitation of general deep learning models.
3. A novel scene understanding module is proposed to promote the integration of the identified vehicle parts and orientation scenes. It also addresses the issues of front-back and left-right category errors and missed detections. Overall, this module contributes positively to the solution to the problem of detecting all vehicle parts.

2. Related work

2.1. Image classification

Image classification is a fundamental problem in computer vision, to classify images into specific categories through computer analysis and processing. Various image classification methods exist, including traditional hand-crafted feature-based methods and end-to-end training using deep learning. In recent years, attention mechanisms, generative adversarial networks, and meta-learning have further boosted the detection capability of image classification models.

Deep learning models have been widely applied to various datasets. Starting from the CNN model AlexNet (Krizhevsky et al., 2012), the field has progressed to the VGG convolutional neural network (Simonyan and Zisserman, 2014). The introduction of residual blocks in deep convolutional neural networks with models like ResNet effectively addresses the gradient vanishing problem. Furthermore, the Inception model (Szegedy et al., 2016), incorporating multi-scale convolutions, and the EfficientNet (Tan and Le, 2019), based on network architecture search, have further advanced the field. Recent research has focused on developing lightweight models, model compression, and model acceleration methods. MobileNet (Howard et al., 2017) and ShuffleNet (Zhang et al., 2018) are some examples of lightweight convolutional neural network models.

Given the inherent relationship between the orientation scene of a vehicle and its parts, exploiting the identification of the corresponding vehicle orientation scene using an image classification model can serve as valuable prior knowledge for accurately detecting vehicle parts. This approach has the potential to enhance the detection accuracy of vehicle parts. Nevertheless, the image classification and detection process for vehicle orientation scenes is often hampered by challenging factors such as lighting conditions, shadows, and mirror-symmetric scenes, which can introduce confusion (Su et al., 2017; Shu et al., 2018; Zhu et al., 2017). To address these limitations, this study applies an improved ResNet image classification model that offers superior feature extraction capabilities, thus enhancing the overall detection performance of vehicle orientation scenes.

2.2. Instance segmentation

Instance segmentation is an essential task in computer vision, aiming to classify and locate each object in an image at the pixel level. The R-CNN series is the earliest set of instance segmentation methods. Its fundamental idea is to extract candidate boxes through the RPN, then classify and regress each candidate box, followed by an ROI-pooling operation based on Spatial Transform Network (STN) to obtain

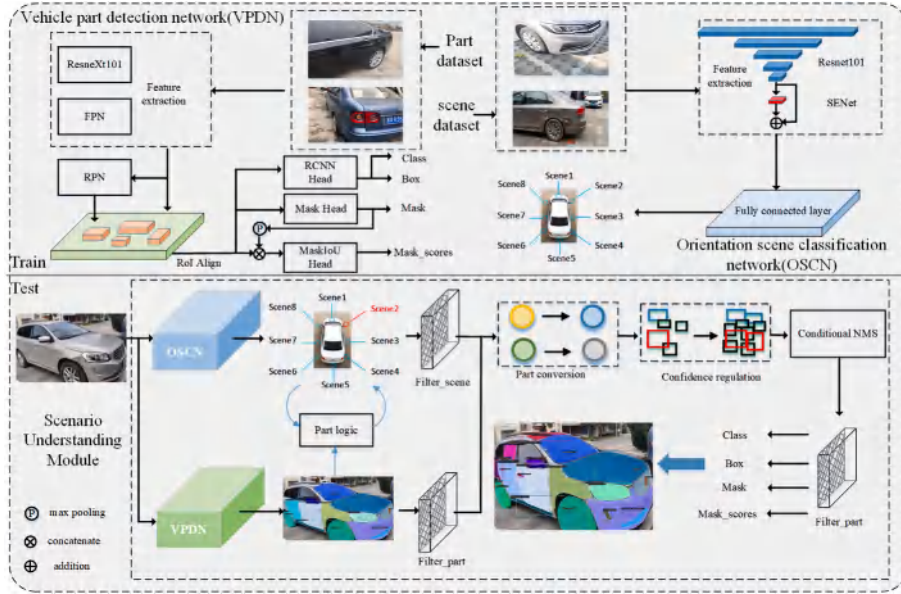


Fig. 1. Overall network structure diagram.

a fixed-size feature map. This process enables final classification and segmentation. Representative works include R-CNN (Girshick et al., 2014), Fast R-CNN (Girshick, 2015), and Faster R-CNN (Ren et al., 2015).

Mask R-CNN (He et al., 2017), developed based on Faster R-CNN, adds a branch network to perform pixel-level segmentation for each Region of Interest (ROI), achieving instance segmentation. Mask R-CNN is one of the most common methods in instance segmentation. Adding a Mask intersection over union (IoU) segmentation detection head network to Mask R-CNN improves performance, resulting in Mask Score R-CNN (Huang et al., 2019). Cascade Mask R-CNN (Huang et al., 2019) adopts a cascaded structure to refine the object detection results iteratively, offering a solution to the challenging issues of distinguishing small objects or objects with inadequate bounding box annotations. This cascaded architecture enhances detection accuracy by effectively filtering out false positives and improving the model's overall robustness. Htc (Chen et al., 2019) has demonstrated remarkable success in enhancing the performance of object detection and instance segmentation by leveraging multi-task learning and incorporating a cascade structure design. In contrast, traditional image segmentation models often fail to capture boundary information and accurately handle intricate structures. Addressing this limitation, Point Rend (Kirillov et al., 2020) introduces rendering ideas into the realm of image segmentation to capture crucial boundaries and fine-grained details better. As a result, PointRend excels in efficiently and precisely characterizing image boundaries and intricate features.

Besides conventional anchor box ideas, QueryInst (Zhang et al., 2021) uses learnable query points instead of anchor boxes to achieve better performance. It reduces the number of hyperparameters needed and improves prediction speed. Instance segmentation research is still evolving, with innovative methods emerging. For example, the recently proposed SpineNet (Du et al., 2021) further enhances the performance of instance segmentation tasks by introducing adaptive multi-scale feature fusion and cross-layer parameter-sharing technologies.

However, more than deep learning models are needed to solve the problem of misclassifications and missed detections due to similar features between categories (Geirhos et al., 2019; Deng et al., 2009). Therefore, this paper proposes a scene-understanding method for vehicle part detection to improve the detection performance of vehicle parts.

3. Research method

This paper proposes an overall network architecture consisting of three parts: the vehicle orientation scene classification (VOSC) sub-network, vehicle part instance segmentation (VPIS) subnetwork, and scene understanding (SU) module. Fig. 1 illustrates the overall network architecture. The subnetworks VOSC and VPIS are present at the upper level, while the lower level comprises the SU module.

In VOSC, we employ transfer learning to input the vehicle orientation scene image and obtain corresponding scene information through the Orientation Scene Classification Network (OSCN), composed of ResNeXt101 and Squeeze-and-Excitation Networks (SENet) (Hu et al., 2018). In VPIS, the vehicle part image is the first input and feature extraction is performed using the ResNeXt101 (Xie et al., 2017) backbone network. Then, basic feature maps are obtained through the RPN, and RoI Align units for object box and class prediction. A fully convolutional network is also utilized to get mask scores from the basic feature maps. The fundamental framework of VPIS comprises the vehicle part detection network (VPDN). In SU module, vehicle orientation scene information obtained from the orientation scene classification network and preliminary results from the instance segmentation network are input to the part logic correction module, enhancing the accuracy and quality of vehicle orientation scene information. This information guides the output of the instance segmentation network through part replacement, confidence regulation, and hierarchical filtering modules to achieve vehicle part classification, object box localization, and segmentation tasks.

3.1. Vehicle orientation scene classification

We conducted in-depth research on the rules of vehicle part scenarios, considering the risk control environment of vehicle part photography angles. Consequently, we proposed the concept of eight-orientation scenes for vehicle parts, as shown in Fig. 2. Vehicle orientation is commonly classified into four cardinal directions: front, back, left, and right. Nevertheless, when the vehicle's orientation falls between two of these cardinal directions, such as at a 45-degree angle, assigning it to a specific category becomes challenging. Moreover, the significant presence of images in the vehicle parts dataset depicting orientations between two cardinal directions contributes to the deficiency in accurate vehicle orientation detection. To address this limitation, a

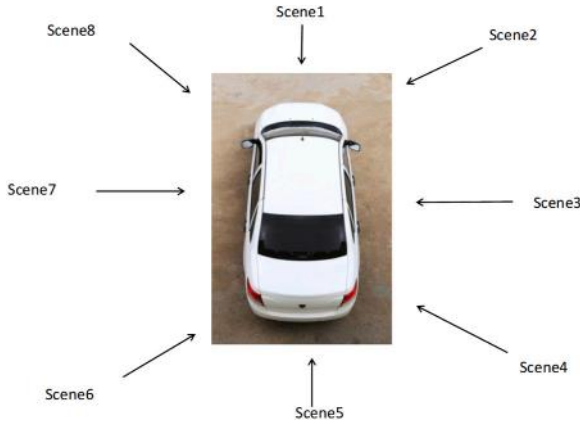


Fig. 2. Eight orientation scene.



Fig. 3. Vehicle orientation scene picture and vehicle parts picture: (a) Pictures of vehicle orientation scene; (b) Pictures of vehicle parts.

more granular approach involves the classification of orientation into 16 directions, with intervals of 22.5 degrees between each direction. However, the closely related characteristics of adjacent directions give rise to substantial mutual influence, ultimately undermining the overall accuracy of orientation detection. As a result, the adoption of eight bits for orientation classification represents a prudent choice, striking an optimal balance between detecting precision and computational efficiency.

We used an improved ResNet101 on the vehicle orientation scene dataset to train the orientation scene classification network. Table 1 shows the structure of the orientation scene classification network, and Fig. 3(a) displays the vehicle orientation scene data.

ResNet convolutional neural network (CNN) was first proposed in 2015, and ResNet101 is one of the multiple-category algorithms of ResNet, an ultra-deep network with 101 layers. The previous CNNs had problems with gradient vanishing and exploding and were only capable of shallow convolutional operations. ResNet networks achieved ultra-deep network structures by proposing residual modules based on the inherent “convolutional layer + pooling layer” module, which allows gradients to be directly passed to the convolutional layer input position during backpropagation through residual modules. ResNet successfully addresses the common problems of gradient vanishing and exploding in CNNs by avoiding error backpropagation calculation in the convolutional layer within the residual module.

While ResNet101 achieved improved network classification accuracy through ultra-deep convolutional operations, the quality of the data images obtained in the field could be better due to low pixel

Table 1
Orientation scene classification network.

Level	Output size	Convolution kernel
conv1	112 × 112	7 × 7, 64, stride 2
conv2_x	56 × 56	3 × 3 max pooling, stride 2
		1 × 1, 64
		3 × 3, 64
conv3_x	28 × 28	1 × 1, 256
		SENet
		× 3
conv4_x	14 × 14	1 × 1, 128
		3 × 3, 128
		1 × 1, 512
conv5_x	7 × 7	SENet
		1 × 1, 256
		3 × 3, 256
		1 × 1, 1024
		SENet
		× 23
		1 × 1, 512
		3 × 3, 512
		1 × 1, 2048
		SENet
		× 3
	1 × 1	Average pool, Fc, Sigmoid
Model calculation force	3.8 × 10 ⁸	

Note: SENet = (Average pool, Fc, Relu, Fc, Sigmoid).

resolutions, object occlusion, and poor lighting conditions, which can affect the recognition performance of the model. To alleviate the interference of these factors on network classification, the SENet attention mechanism was introduced in the ResNet101 network. The SENet module consists of an Average pool, Fc, ReLU, Fc, and Sigmoid and is embedded in each basic residual block unit of ResNet, as shown in Fig. 4. The *Scale* operation depicted in Fig. 4 is employed to extend the output generated by the sigmoid activation function, ensuring it matches the exact dimensions of the input feature map tensor. Subsequently, the two tensors are element-wise multiplied in order to derive the ultimate output tensor. By assigning different weights to channels in the feature map, with higher weights given to channels associated with foreground objects and lower weights assigned to background-related channels, the SENet module strengthens the convolutional operations on channels associated with foreground objects. It increases the feature representation ability of the model, thereby improving recognition accuracy. ResNet101 and SENet serve as the backbone networks for the OSCN, offering notable advantages in terms of simplicity and efficiency. These characteristics guarantee enhanced performance in orientation scene detection while facilitating seamless integration into the scene understanding module.

The input of OSCN consists of images from the VOD. By leveraging the ResNet101 and SENet, we obtain a high-dimensional feature map F_{OSCN} , $F_{OSCN} \in R^{W_1 \times H_1 \times D_1}$, where W_1 and H_1 are the width and height of the feature map, and D_1 is the number of channels in the feature map. The formula of F_{OSCN} is as follows:

$$F_{OSCN} = CNN_{ResNet101+SENet}(I_{VOD}, \theta_1) \quad (1)$$

where I_{VOD} is the input image of VOD and θ_1 is the learnable parameter in the $CNN_{ResNet101+SENet}$ model.

Then, F_{OSCN} undergoes the Average pooling module, a fully connected layer and sigmoid activation function to attain the eight scene scores $Scene_{scores}$, $Scene_{scores} \in R^{1 \times 8}$, as shown in Eq. (2). $Scene_{scores}$ represents the output scores for each of the eight scene categories.

$$Scene_{scores} = \text{Sigmoid} [FC (Average_pooling(F_{OSCN}))] \quad (2)$$

Finally, $Scene_{scores}$ are processed through the Argmax function to output the possible scene category $Scene$, $Scene \in R^{1 \times 1}$, as shown in Eq. (3). The $Scene$ is the final output of the OSCN model.

$$Scene = \text{Arg max} (Scene_{scores}) \quad (3)$$

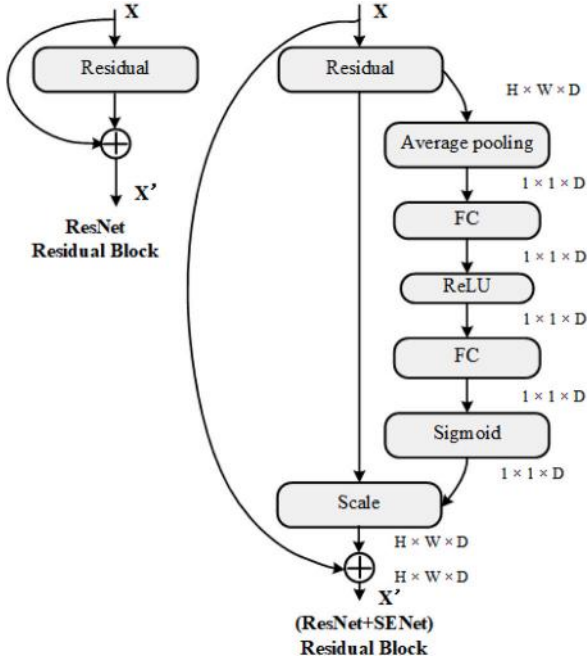


Fig. 4. The residual block difference between ResNet and (ResNet + SENet).

3.2. Vehicle part instance segmentation

The vehicle part images, as shown in Fig. 3(b), consist of 59 different part types to be identified, as listed in Table 2, covering the vast majority of parts in the vehicle. The Mask Scoring R-CNN model, as shown in the VPDN section of Fig. 1, was used for instance segmentation. Mask Scoring R-CNN tackles the issue of inaccurate instance segmentation by introducing an additional mask scoring branch. This branch enables the model to assign a confidence score to each predicted segmentation mask, effectively quantifying the quality of the segmentation. By incorporating this mask scoring branch, Mask Scoring R-CNN can offer more dependable and precise instance segmentation outcomes compared to other models. Moreover, this model demonstrates exceptional proficiency in capturing fine-grained details and accurately delineating object boundaries. It is particularly relevant in applications such as vehicle part detection and the fusion of scene understanding modules.

The input of VPDN consists of images from the VMPD. By leveraging the ResNeXt101 and Feature Pyramid Network (FPN) (Lin et al., 2017) feature extraction network, we obtain a high-dimensional feature map F , $F \in R^{W_2 \times H_2 \times D_2}$, where W_2 and H_2 are the width and height of the feature map, and D_2 is the number of channels in the feature map. Fig. 5 elucidates the disparity in the residual block approach employed by ResNet and ResNeXt. It is imperative to note that ResNeXt-101 and ResNet-101 diverge mainly in terms of their block structures and the strategy employed for parameter sharing. ResNeXt incorporates parallel convolution pathways that effectively leverage parameter sharing, yielding a streamlined and efficient parameter representation. The architecture of FPN is shown in Fig. 6, which comprises three crucial components: the Bottom-up pathway, the Top-down pathway, and the Lateral connections. In the Bottom-up feature extraction network, lower-level features are extracted and processed through a convolutional network, resulting in a higher-resolution feature pyramid. Subsequently, the Top-down feature transfer network employs up-sampling operations to restore the features subsampled by higher-level semantic features back to their original resolution. This process is repeated to ensure complete restoration. Finally, the fusion of higher-resolution features with their lower-resolution counterparts occurs through the Lateral connections, leveraging the combined

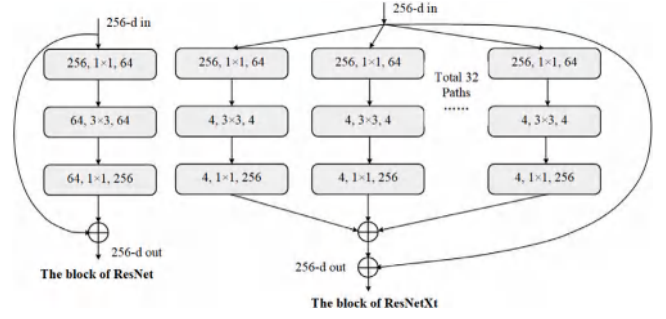


Fig. 5. The block difference between ResNet and ResNeXt.

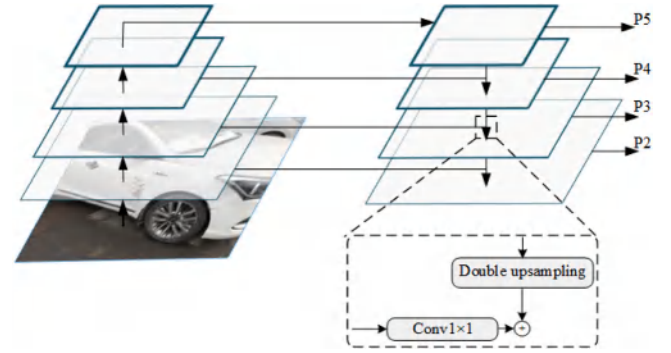


Fig. 6. FPN structure.

advantage of semantic and detailed information. The horizontal connections facilitate the exchange of information between different levels, thereby generating multi-scale feature maps with a wealth of semantic expression capabilities. FPN demonstrates remarkable capabilities in accurately detecting objects of diverse scales, thereby significantly enhancing detection accuracy. The formula of F is as follows:

$$F = CNN_{ResNeXt101+FPN}(I_{VPDN}, \theta_2) \quad (4)$$

where I_{VPDN} is the input image of VPDN and θ_2 is the learnable parameter in the $CNN_{ResNeXt101+FPN}$ model.

Then, F undergoes the Region Proposal Network (RPN) module to extract the region proposal boxes F_p , $F_p \in R^{N_1 \times 4}$, as well as the classification score RPN_{cls} , $RPN_{cls} \in R^{N_2 \times 2k}$, associated with each region proposal box by using several anchor boxes. N_2 represents the total number of region proposal boxes, k represents the number of anchor boxes, where each region proposal box is associated with k anchor boxes. Each anchor box is characterized by two classification scores: one for the foreground object and the other for the background. F_p comprises the boundary coordinates of each region proposal box, specifically the lower-left corner (x_1, y_1) and the upper-right corner (x_2, y_2). The RPN leverages the convolutional layer to predict the classification score RPN_{cls} . Based on RPN_{cls} , the anchor boxes are filtered, and only N_1 region proposal boxes with high scores are retained.

$$[F_p, RPN_{cls}] = RPN(F) \quad (5)$$

The F_p and F are subjected to Region of Interest Align (RoI Align) to achieve feature alignment between F_p and F , resulting in F_{RoI} , the aligned region of interest features. F_{RoI} are then fed into the head detection network. Specifically, $F_{RoI(7 \times 7)}$ ($F_{RoI(7 \times 7)} \in R^{N_1 \times D_2 \times 7 \times 7}$) and $F_{RoI(14 \times 14)}$ ($F_{RoI(14 \times 14)} \in R^{N_1 \times D_3 \times 14 \times 14}$) represent the region size of RoI Align, which is set as 7×7 and 14×14 , respectively. The choice of these region sizes is motivated by the specific tasks performed by different heads in the network architecture. In the $RCNN_{Head}$, the primary tasks are object classification and bounding box regression, for which a region size of 7×7 is sufficient to capture the relevant features

Table 2
Vehicle part class.

Part category			
1. Roof_outer_panel	2. Outer_mirror(right)	3. Outer_mirror(left)	4. Mirror_cover(right)
5. Mirror_cover(left)	6. Bottom_edge(right)	7. Bottom_edge(left)	8. Steel_ring
9. Baggage_cover	10. Rear_bumper_electric_eye	11. Rear_bumper_skin	12. Rear_bumper_decorative_light(right)
13. Rear_bumper_decorative_light(left)	14. Rear_window_glass	15. Rear_door_glass(right)	16. Rear_door_glass(left)
17. Back_door_shell(right)	18. Back_door_shell(left)	19. Rear_door_trim(right)	20. Rear_door_trim(left)
21. Back_door_handle(right)	22. Back_door_handle(left)	23. Rear_fender(right)	24. Rear_fender(left)
25. Rear_fender_wheel_eyebrow(right)	26. Rear_fender_wheel_eyebrow(left)	27. liftgate_glass	28. liftgate_shell
29. Tire	30. Inner_tail_light(right)	31. Inner_tail_light(left)	32. Front_bumper_skin
33. Front_bumper_lower_grille	34. Head_lamp(right)	35. Head_lamp(left)	36. Front_window_glass
37. Front_door_glass(right)	38. Front_door_glass(left)	39. Car_right_door	40. Car_left_door
41. Front_door_trim(right)	42. Front_door_trim(left)	43. Front_door_handle(right)	44. Front_door_handle(left)
45. Fog_lamp(right)	46. Fog_lamp(left)	47. Front_fender(right)	48. Front_fender(left)
49. Front_fender_wheel_eyebrow(right)	50. Front_fender_wheel_eyebrow(left)	51. Exterior_tail_light(right)	52. Exterior_tail_light(left)
53. Tail_light(right)	54. Tail_light(left)	55. Fuel_tank_cap	56. Grille
57. Grille_logo	58. Engine_cover	59. License_plate	

of the object. On the other hand, in the *Mask_Head*, more detailed predictions of the pixel-level masks are required, necessitating a larger region size of 14×14 to extract richer and more comprehensive feature information.

$$F_{RoI(7 \times 7)} = RoI_{Align(7 \times 7)}(F_p, F) \quad (6)$$

$$F_{RoI(14 \times 14)} = RoI_{Align(14 \times 14)}(F_p, F) \quad (7)$$

Then, $F_{RoI(7 \times 7)}$ is fed into *RCNN_Head* to obtain the predicted class probabilities ($Class_{final}, Class_{final} \in R^{N_1 \times 7 \times (C_1+1)}$) and bounding box adjustments ($Box_{final}, Box_{final} \in R^{N_1 \times 4}$). Simultaneously, $F_{RoI(14 \times 14)}$ is passed through *Mask_Head* to generate the predicted masks ($Mask_{final}, Mask_{final} \in R^{N_1 \times 1 \times M \times M}$). '1' indicates the number of channels and M indicates the size of the mask. Each mask prediction is a binary matrix representing the mask shape of the instance corresponding to RoI. The $Mask_{final}$, along with the $F_{RoI(14 \times 14)}$, are then forwarded to the *MaskIoU_Head* to compute the mask scores ($Maskscore_{final}, Maskscore_{final} \in R^{N_1}$). $Maskscore_{final}$ are used to evaluate the degree of overlap between the mask of RoI and the real mask.

$$[Class_{final}, Box_{final}] = RCNN_Head(F_{RoI(7 \times 7)}) \quad (8)$$

$$Mask_{final} = Mask_Head(F_{RoI(14 \times 14)}) \quad (9)$$

$$Maskscore_{final} = MaskIoU_Head(Addition(Maxpooling(Mask), F_{RoI})) \quad (10)$$

3.3. Scene understanding module

Certain detection results were achieved after training the vehicle parts dataset using the Mask Scoring R-CNN model. However, among the 59 types, there are part categories that include front and back as well as left and right, such as 'Outer_mirror(right)', 'Outer_mirror(left)', 'Rear_door_glass(right)', 'Front_door_glass(right)', etc. These parts have remarkably similar features and thus are easily misclassified. Furthermore, many instances of missed detections and overlapping detections were observed in the tested images.

Therefore, a scene-understanding method for vehicle multi-part detection under risk control was proposed, which combines the orientation scene image classification model with the conventional vehicle part instance segmentation model. This approach partially solves the problems of misclassifications caused by the similarity of features among categories, as well as missed detections and overlapping detections. The fusion method involves using a multifunctional module: the scene understanding module. The scene understanding module consists of four functional modules: the part logic correction module, part replacement module, confidence regulation module, and hierarchical filtering module.

3.3.1. Part logic correction module

Influence from lighting, shadows, and mirror symmetry in the image can make scene recognition models clear, resulting in misclassifications. The part logic correction module is introduced to correct the scene when it deviates from the actual position detected by the orientation scene classification network. Due to the left-right symmetry of vehicle parts, using a deep learning model to identify parts with both left and right categories, such as Outer_mirror(right) and Outer_mirror(left), often results in detection of both categories, which interferes with accurate detection due to their similar features. Deep learning models are not effective at detecting such cases. However, the positions of the two category boxes are generally similar, making the part logic correction module possible. Using the relationship between various vehicle parts, we refine the scene to complete the part recognition and detection task. A few key parts used in part logic are displayed in Fig. 7.

For example, the scene can be fine-tuned by comparing the horizontal coordinates of the center point of the bounding box of objects 47, 48 (Front_fender) and 2, 3, 4, 5 (Outer_mirror, Mirror_cover). If the horizontal coordinate of the center point of the Front_fender bounding box is greater than that of the Outer_mirror and Mirror_cover's bounding box, the scene is adjusted to Scene 2. If not, it is adjusted to Scene 8. The logical relationships among multiple parts adjust the scene, which can significantly ensure its correctness. Adjusting close-range pictures with fewer parts using the part logic correction method is not feasible. However, the orientation scene classification network can achieve better results. Combining this network with part logic correction can significantly improve the scene detection effect for medium and close-range scenes.

After using the part logic correction module, the detected scenes can be correctly recognized, and the corresponding vehicle parts summarized from the correct vehicle scenes can play a role in other modules of scene understanding that follow.

3.3.2. Part replacement module

After obtaining the correct vehicle scene information corresponding to the image, we import the pre-calculated vehicle part information for each vehicle scene and apply the part replacement module, as shown in Fig. 8. The vehicle orientation scene information is then used to guide the instance segmentation network in classifying its output categories. The interconversion between the left and right categories, as well as the front and rear categories, facilitates the preservation of scene-specific categories. This drives the transformation of the left and right instance boxes, as well as the front and rear instance boxes. However, instance boxes unrelated to the front, rear, left and right categories remain unaltered. This process is demonstrated in Eq. (11) as follows:

$$Labels_{scene} = \begin{cases} Labels_{scene}, & \text{if } Labels \text{ not in } Scene \\ Labels, & \text{if } Labels \text{ in } Scene \end{cases} \quad (11)$$

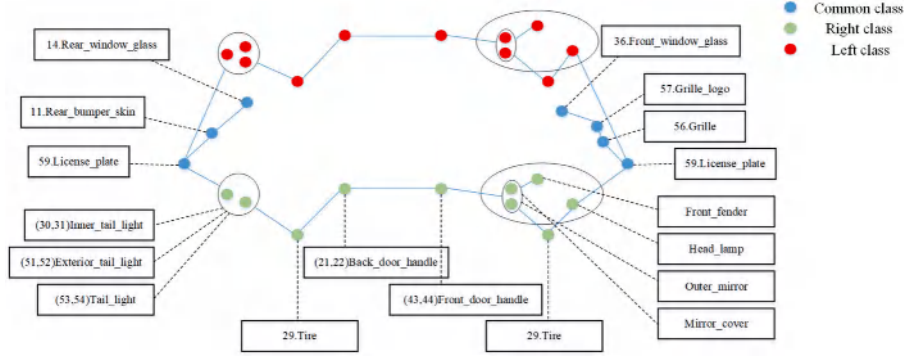


Fig. 7. Part logic correction module.

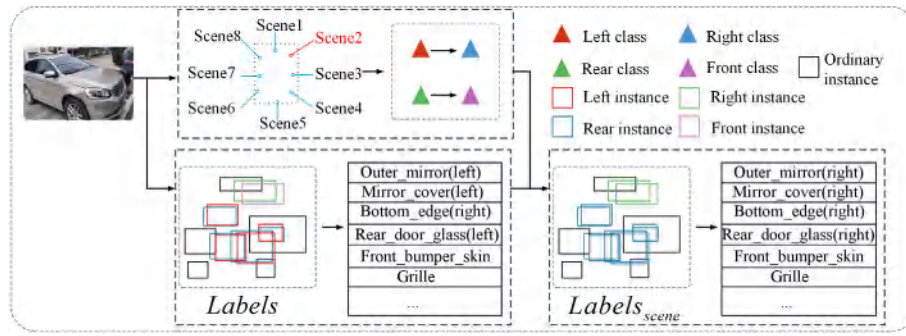


Fig. 8. Part replacement module.

The pre-calculated vehicle part information for each vehicle scene plays a crucial role in the equation.

Due to multiple front-back and left-right distinctions within the 59 categories of the vehicle part dataset, similar features between these categories can often lead to false detections, thereby reducing the precision of the detection. Adding the part replacement module effectively improves the model's recognition accuracy for these categories.

3.3.3. Confidence regulation module

The vehicle part replacement module can significantly improve the accuracy of vehicle recognition. However, it does not affect the confidence scores of the categories, and thus the original problem of missed detections still exists. To address this issue, we propose a confidence regulation module.

The confidence regulation module first performs a statistical analysis of the detection results of the instance segmentation model, identifying the categories likely to be missed in various scenes. For the categories obtained after applying the part replacement module to Eq. (11), confidence regulation is then conducted using Eq. (12).

$$Scores_{scene} = \begin{cases} Scores + conf, & \text{if (Labels in Scene) and } (Scores \geq \alpha) \\ Scores, & \text{else} \end{cases} \quad (12)$$

In Eq. (12), the parameter α is a low-dimensional parameter for the first-level filtering objective mechanism, with an amplitude ranging from 0 to 0.1, while $conf$ is the confidence regulation parameter.

Some detection effectiveness can be achieved after training the instance segmentation model on the vehicle part dataset. Still, due to the dataset's poor quality and inaccurate labeling, there are often missed detections during the detection process. The confidence regulation module can significantly improve the problem of missed detections under high-quality confidence score thresholds.

3.3.4. Hierarchical filtering module

To further address the issues of missed detections and misclassifications, after performing low-dimensional hierarchical filtering object, part replacement driven by the vehicle orientation scene and logic correction module, and confidence regulation, the results obtained from Eqs. (11) and (12) need to undergo high-dimensional hierarchical filtering object, as shown in Eqs. (13) and (14):

$$Scores_{final} = \begin{cases} Scores_{scene}, & \text{if } Scores_{scene} \geq score_{thr} \\ delete, & \text{else} \end{cases} \quad (13)$$

$$Labels_{final} = \begin{cases} Labels_{scene}, & \text{if } Scores_{scene} \text{ in } Scores_{final} \\ delete, & \text{else} \end{cases} \quad (14)$$

In Eq. (13), $score_{thr}$ is the high-dimensional confidence threshold, with an amplitude ranging from 0.3 to 0.7, and $Scores_{final}$ represents the final confidence score array of the obtained objects, used to conditionally filter out the last part categories $Labels_{final}$. The scene module's confidence regulation and hierarchical filtering mechanisms can alleviate the problem of missed detections under high-quality confidence score threshold conditions.

After passing through the modules mentioned above, duplicate bounding boxes of the same detected object may still exist. To address this issue, we introduce the Conditional Non-Maximum Suppression module (ConditionalNMS) after advanced hierarchical filtering, as shown in Eq. (15):

$$(Labels_{final}, Scores_{final}) = \text{ConditionalNMS}(Labels_{final}, Scores_{final}, NMS_{iou}) \quad (15)$$

In Eq. (15), ConditionalNMS refers to the conditional non-maximum suppression unit, which differs from traditional NMS in that it only performs NMS operations on the categories of objects that have undergone part replacement. NMS_{iou} is the IoU threshold for deleting overlapping objects. When the IoU ratio between a high-confidence object box

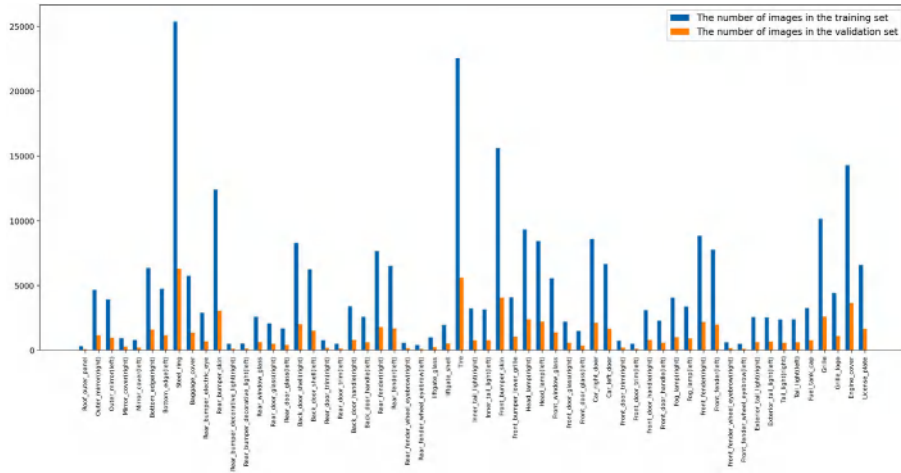


Fig. 9. VMPD images distribution.

and a low-confidence object box is more extensive than NMS_{iou} , the lower-confidence object box is removed. ConditionalNMS can address the problem of overlapping bounding boxes after part replacement, thereby improving the detection performance.

4. Experimental results and analysis

4.1. Data set source

In this study, we selected 59 categories of vehicle parts as experimental objects, as shown in Table 2. To ensure the quality and reliability of the data, extensive data cleaning and error labeling procedures were executed. We used the established COCO dataset construction methodology to curate two distinct vehicle component datasets. An insurance company generously provided the image dataset used in this study. The first dataset is VOD, a small dataset consisting of eight scenes required by the image classification model for vehicle orientation scenes, as illustrated in Fig. 2. The training and test set samples in VOD are 39,058 and 4335 images, respectively, with a ratio of 9:1. The second dataset, VMPD, is a large vehicle part dataset required by the instance segmentation model for vehicle part detection. The training and test samples in VMPD are 45,503 and 11,376 images, respectively, with a ratio of 3:1. The dataset contains 59 categories of objects, with 287,331 and 71,712 instances, respectively. The datasets are not publicly available, and the distributions of the data sets are visually represented in Figs. 9, 10, and 11. Fig. 9 illustrates the distribution of images across various VMPD categories, while Fig. 10 showcases the distribution of target boxes corresponding to different VMPD categories. Notably, the number of target boxes in each category resembles the number of images, indicating a consistent occurrence frequency for each category within the VMPD images. This observation aligns with the objective nature of VMPD, predominantly capturing images of vehicles. Additionally, Fig. 11 presents the distribution of images within various categories in VOD. Notably, each category exhibits many images except for *Scene3* and *Scene7*. This trend is consistent with the fact that vehicle collision accidents predominantly occur near the front and rear of the vehicle.

4.2. Experimental environment

The model described in this paper was trained and tested using the NVIDIA 1080Ti professional acceleration card. The operating system was Ubuntu16.04.6 LTS, and CUDA11.1 was used for training acceleration. Python 3.8 was used as the programming language, and the Pytorch network development framework was used. During the training phase of the VOD dataset, the batch size was set to 16, and

the stochastic gradient descent (SGD) algorithm was used. The initial learning rate was set to 0.0001, and the learning rate was reduced by a factor of 10% every 9 epochs, with a maximum of 100 epochs. In the VMPD dataset, we used the multi-GPU training method, set the batch size to 2, and used the SGD algorithm. The initial learning rate was 0.001, and the learning rate was reduced by a factor of 10% every nine epochs, with a maximum of 36 epochs. The *Scene_thr* value for the part logic correction module is set to 0.7. For the first-stage low-dimensional parameter α in the hierarchical filtering mechanism, a value of 0.05 is chosen. Furthermore, the high-dimensional confidence threshold *Score_thr* for the filtering module is set to 0.5 to facilitate the accurate selection of relevant object candidates. The ConditionalNMS module with an IoU threshold (NMS_{iou}) value of 0.65 is employed to address the issue of duplicate bounding boxes for the same object after undergoing part replacement. By using GPUs, this method can be easily extended to a distributed system.

4.3. Evaluation index

In this paper, we adopt the widely used evaluation metrics, Average Precision (AP) and mean Average Precision (*mAP*) (He et al., 2017; Tan et al., 2020; He et al., 2016), to comprehensively evaluate the performance of the object detection and instance segmentation models. The AP and *mAP* is defined as follows:

$$AP(n) = \int_0^1 P_n(R_n) dR_n \quad (16)$$

$$P_n = \frac{TP(n)}{TP(n) + FP(n)} \quad (17)$$

$$R_n = \frac{TP(n)}{TP(n) + FN(n)} \quad (18)$$

$$mAP = \frac{1}{c_1} \sum_{n=1}^{c_1} AP(n) \quad (19)$$

In Eq. (16), AP is defined as the value obtained by calculating Precision (P) and averaging it across different confidence thresholds. It serves as a measure of the model's accuracy under varying confidence thresholds, where the confidence threshold is denoted as $Score_{final}$. $AP(n)$ represents the average detection accuracy of category n , while R_n indicates the recall rate of category n , and P_n represents the detection accuracy of category n at a specific $Score_{final}$. $P_n(R_n)$ specifically denotes the precision of category n when the recall rate of category n is R_n , which is obtained under $Score_{final}$. The calculation formulas for R_n and P_n are presented in Eq. (17) and (18), respectively. In these equations, $TP(n)$, $FP(n)$, and $FN(n)$ represent the true positive (correctly predicting objects of category n as category n), false positive

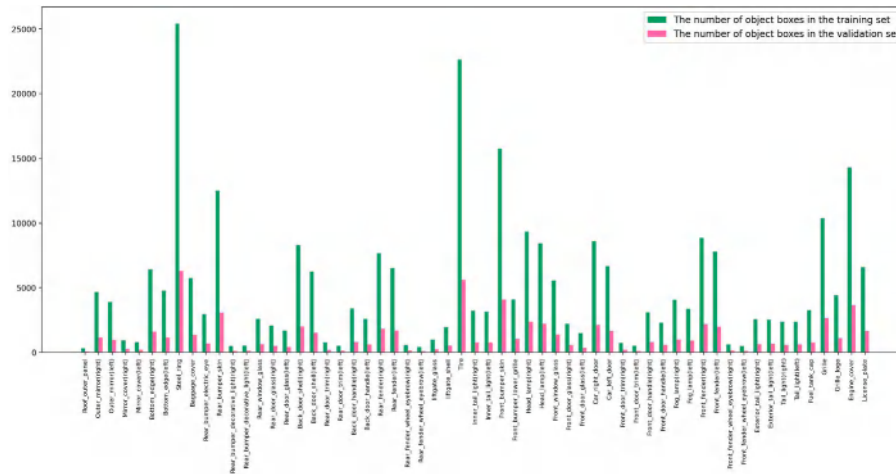


Fig. 10. VMPD target boxes distribution.

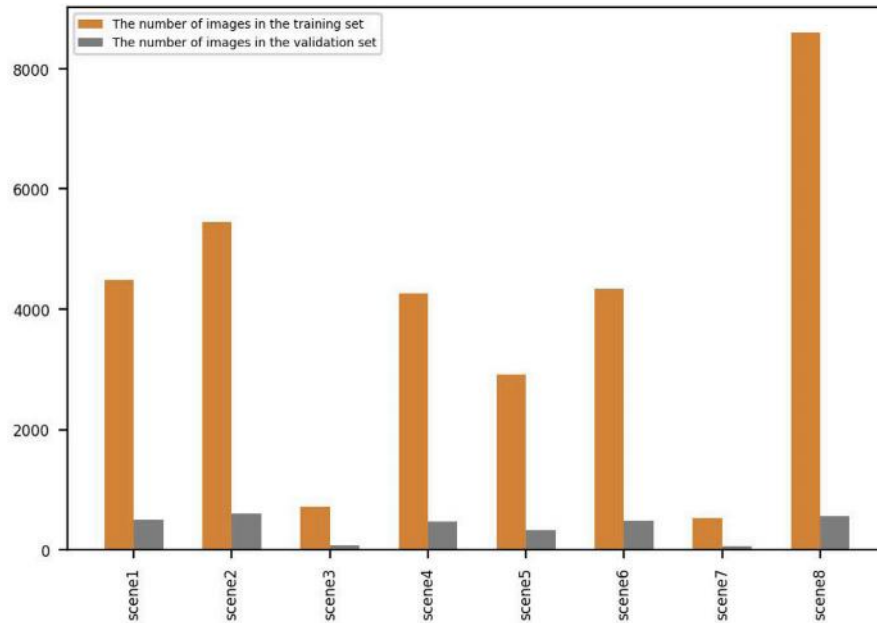


Fig. 11. VOD data set distribution.

Table 3
Results of orientation scenario and part logic module ablation experiment.

Method	Scene	Part logic correction	Part replacement	Hierarchical screening	$mAP_b@50$ (%)	$mAP_s@50$ (%)
MS R-CNN			✓	✓	54.1	53.2
MS R-CNN	✓		✓	✓	58.4	57.7
MS R-CNN	✓	✓	✓	✓	66.3	65.5

(incorrectly predicting objects of other categories as category n), and false negative (incorrectly predicting objects of category n as other categories) results, respectively. In Eq. (19), C_1 represents the number of categories of vehicle parts. The mAP is employed to quantitatively evaluate the training effect of the model, which provides an assessment of the average detection accuracy across all categories and the overall performance of the training model.

4.4. Ablation experiment

We conducted the following ablation studies to investigate the impact of integrating the vehicle orientation scene and part logic correction module.

The $mAP_b@50(\%)$ and $mAP_s@50(\%)$ represent the mean Average Precision of bounding box results and instance segmentation results,

Table 4

Results of the ablation experiment of the confidence regulation module.

Method	Scene	Part logic correction	Part replacement	Confidence control	Hierarchical screening	$mAP_b@50$ (%)	$mAP_s@50$ (%)
MS R-CNN	✓	✓	✓		✓	43.3	43.4
MS R-CNN	✓	✓	✓	✓	✓	49.5	51.9

Table 5

Results of orientation scenario and part logic module ablation experiment.

Method	Scene	Part logic correction	Part replacement	Hierarchical screening	$mAP_b@50$ (%)	$mAP_s@50$ (%)
MS R-CNN	✓	✓			54.1	53.2
MS R-CNN	✓	✓	✓		62.4	61.0
MS R-CNN	✓	✓	✓	✓	66.3	65.5

respectively, when the IoU overlap threshold is set to 0.5, and the confidence score threshold is set to 0.05, which is consistent with the parameters used during training. Table 3 shows a slight improvement in detection performance after introducing the vehicle orientation scene, part replacement, and hierarchical filtering modules. The best detection results were obtained after adding the part logic correction module.

To investigate the impact of the confidence regulation module, we conducted the following ablation experiment with a confidence score threshold set to 0.5. The results in Table 4 demonstrate that the confidence regulation module can significantly enhance the detection performance of vehicle parts in instance segmentation, especially under high confidence score thresholds. Finally, we investigate the effects of the part replacement and hierarchical filtering modules by conducting the following ablation experiment.

The results in Table 5 demonstrate that the simple introduction of the scene or part logic correction module provides information about the part category in the corresponding scene but has a little direct impact on the detection results. Only when the scene understanding is fused with the part replacement and filtering modules can the scene introduction and part logic correction modules enhance the detection performance. When only the part replacement module is integrated, some improvement in detection accuracy can be achieved because the part replacement combined with the correct scene can improve the misclassifications of left-right and front-back categories of vehicle parts. However, the best results are achieved when the part replacement module and the hierarchical selection module are integrated because the hierarchical filtering module can alleviate the issue of excessive detection by the baseline model, thereby improving the accuracy of part detection.

4.5. Comparison experiment of model detection results

4.5.1. Orientation scene comparison experiment

There are many ways to incorporate the vehicle orientation scene, including during training or later inference stages. In this study, we incorporated the vehicle orientation scene in the later inference stage. The dataset used to train and validate the OSCN is different from the dataset used for the vehicle part instance segmentation network. Therefore, the OSCN is fused with the VPDN to improve the OSCN's generalization ability in detecting vehicle part instances. For comparison, Table 6 presents the results of the OSCN compared with other advanced image classification models. The evaluation metric employed in this study is the widely used $Top-1$ Accuracy, which measures the proportion of correctly classified images. In $Top-1$ Accuracy, a correctly classified image is defined as one where the predicted category with the highest probability from the model aligns precisely with the ground truth category. This criterion is mathematically expressed as follows:

$$Accuracy(n) = \frac{I_{true}(n)}{I_{total}(n)} \times 100\% \quad (20)$$

$$Accuracy_{total} = \frac{\sum_{n=1}^{C_2} I_{true}(n)}{\sum_{n=1}^{C_2} I_{total}(n)} \times 100\% \quad (21)$$

Table 6

Experimental results of orientation scene network comparison.

Method	$Accuracy_{total}$ (%)
Resnet34	89.8
Resnet101	92.3
Vgg19	26.9
Alexnet	82.1
Mobilenet_v2	61.4
Googlenet	86.2
EfficientNet	72.7
Regnet	85.1
Ours	93.7

In Eqs. (20) and (21), $Accuracy(n)$ denotes the $Top-1$ accuracy of category n , which is calculated as the ratio of accurately predicted images of category n ($I_{true}(n)$) to the total number of images of category n ($I_{total}(n)$). In the context of an image classification model, C_2 represents the total number of categories utilized. The $Accuracy_{total}$ signifies the average $Top-1$ accuracy across all categories of the image classification model. As the table above demonstrates, the OSCN achieves high detection accuracy, providing a solid foundation for subsequent scene fusion.

4.5.2. Orientation scene ablation study

In the OSCN model, the ResNet101 network was used as the baseline model, and the addition of the SENet module improved the accuracy of the vehicle scene classification.

Table 7 shows the results of the ablation experiments for the OSCN model. The results demonstrate that adding the attention module to the baseline model can improve the mean classification accuracy for each category from 92.3% to 93.7%. Specifically, the classification accuracy for seven of the eight orientation categories increased to varying degrees, while the precision for orientation category *Scene2* slightly decreased.

4.5.3. Part instance segmentation comparison experiment

The ultimate goal of this study is to achieve high-precision detection of multiple vehicle parts through instance segmentation. Since insurance reimbursement requires differentiating between left, right, up, and down positions, the vehicle part dataset also makes such differentiations. These can easily lead to misclassifications due to similar features between different parts. We fused the OSCN and VPDN through the scene understanding module to achieve high-precision detection of vehicle part instances. To evaluate the advancement of the proposed model, various instance segmentation models were implemented on the vehicle part dataset and compared, including Mask R-CNN, Cascade Mask R-CNN, MS R-CNN, SOLOv2, QueryInst, Htc and Point rend. The test results, with a confidence score threshold 0.05, are shown in the Tables 8 and 9, with Table 8 showing the overall detection performance of the model and Table 9 showing the accuracy of each category of vehicle parts.

Table 7
Ablation experiment of OSCN model.

Method		Accuracy (%)								Accuracy _{total} (%)
ResNet101	SENet	Scene1	Scene2	Scene3	Scene4	Scene5	Scene6	Scene7	Scene8	—
✓		91.5	94.0	81.0	94.5	88.0	91.3	73.7	95.7	92.3
✓	✓	93.0	93.6	83.5	96.0	90.4	94.4	80.7	96.3	93.7

Table 8
Part comparison test results.

Method	$mAP_b@50$ (%)	$mAP_b@50 : 95$ (%)	$mAP_s@50$ (%)	$mAP_s@50 : 95$ (%)
Mask R-CNN	53.0	43.0	52.7	42.2
Cascade Mask R-CNN	53.7	44.6	52.9	42.5
MS R-CNN	54.1	42.5	53.2	42.6
SOLOv2	—	—	52.4	42.3
QueryInst	28.9	22.7	29.2	23.2
Htc	55.3	46.9	55.3	45.0
Point Rend	52.0	42.4	51.9	42.3
Ours	66.3	51.4	65.5	51.8

Table 9
Experimental results of specific categories of vehicle parts.

Category	$AP_b@50$ (%)						$AP_s@50$ (%)					
	Mask R-CNN	Cascade Mask R-CNN	MS R-CNN	SOLO-v2	Query-Inst	Our model	Mask R-CNN	Cascade Mask R-CNN	MS R-CNN	SOLO-v2	Query-Inst	Our model
Roof_outer_panel	40.1	28.9	36.2	—	12.7	36.2	40.3	29.9	38.6	38.6	12.4	38.6
Outer_mirror(right)	62.2	65.5	62.1	—	32.4	87.4	62.1	66.3	60.1	55.5	32.3	88.1
Outer_mirror(left)	54.0	57.8	54.2	—	33.7	84.4	54.6	57.9	55.4	41.1	34.4	85.4
Mirror_cover(right)	23.8	24.2	20.8	—	6.5	34.0	23.8	24.3	20.6	17.2	9.2	34.8
Mirror_cover(left)	17.5	15.7	18.4	—	8.8	26.8	17.4	15.9	19.2	11.5	10.9	28.0
Bottom_edge(right)	52.8	51.5	52.1	—	23.2	67.7	49.2	42.4	48.9	52.6	22.3	64.4
Bottom_edge(left)	38.5	33.7	33.4	—	20.3	64.0	36.7	28.9	33.6	32.9	19.7	60.3
Steel_ring	97.6	97.5	97.4	—	78.8	97.4	97.6	97.5	96.9	97.3	79.3	96.9
Baggage_cover	94.7	93.9	93.8	—	49.4	93.8	94.5	93.7	92.6	94.4	49.4	92.6
Rear_bumper_electric_eye	41.1	41.0	40.3	—	21.3	40.3	41.6	41.1	41.4	38.9	22.6	41.4
Rear_bumper_skin	96.4	95.3	94.3	—	51.7	94.3	95.3	94.9	93.5	95.5	51.4	93.5
Rear_bumper_decorative_light(right)	28.9	30.3	26.2	—	13.1	26.9	28.9	31.0	27.6	29.8	13.5	27.9
Rear_bumper_decorative_light(left)	23.2	23.8	25.2	—	9.3	28.9	23.2	23.8	24.8	26.6	9.6	28.6
Rear_window_glass	93.6	91.1	93.4	—	49.8	93.4	93.8	91.1	92.5	91.7	49.6	92.5
Rear_door_glass(right)	40.3	40.4	37.6	—	20.4	58.9	40.2	40.1	38.2	36.5	20.4	59.2
Rear_door_glass(left)	30.9	27.6	28.6	—	14.2	58.8	30.8	27.6	28.0	27.0	14.1	58.2
Back_door_shell(right)	56.0	70.9	60.8	—	29.7	81.0	55.9	70.9	63.0	63.2	29.7	81.6
Back_door_shell(left)	44.5	58.5	41.8	—	27.5	78.9	44.5	58.3	40.2	52.8	27.5	77.3
Rear_door_trim(right)	31.5	24.7	30.0	—	9.5	40.4	29.7	21.3	26.7	31.9	9.3	36.9
Rear_door_trim(left)	25.0	19.0	25.5	—	7.2	40.9	23.3	17.1	24.3	17.9	8.4	38.5
Back_door_handle(right)	40.8	50.6	41.5	—	26.5	67.1	40.5	50.7	42.2	48.4	7.0	67.4
Back_door_handle(left)	30.7	38.0	35.5	—	15.7	65.0	30.8	37.7	33.7	32.1	16.1	65.5
Rear_fender(right)	53.1	47.9	53.6	—	23.7	76.6	53.3	48.1	54.8	55.3	24.1	79.2
Rear_fender(left)	51.8	48.3	53.0	—	31.2	78.8	52.0	48.9	51.7	60.2	31.7	79.0
Rear_fender_wheel_eyebrow(right)	46.1	36.6	45.1	—	12.5	55.9	43.2	31.3	36.6	45.4	12.8	42.6
Rear_fender_wheel_eyebrow(left)	22.7	21.8	22.7	—	17.8	48.0	22.3	20.8	21.6	26.0	17.6	36.1
liftgate_glass	77.7	76.1	76.6	—	33.4	76.6	77.7	76.1	75.8	81.4	33.4	75.8
liftgate_shell	85.4	83.2	82.8	—	42.7	82.8	83.9	82.1	82.0	84.1	41.0	82.0
Tire	87.6	87.8	87.2	—	62.6	87.2	87.4	87.5	86.7	85.7	62.7	86.7
Inner_tail_light(right)	47.8	64.1	63.1	—	25.6	57.2	47.8	63.8	67.6	53.5	25.7	64.7
Inner_tail_light(left)	52.1	73.9	84.2	—	38.2	51.9	52.1	73.9	78.3	51.8	38.3	52.7
Front_bumper_skin	97.1	96.0	96.4	—	55.6	96.4	96.2	95.5	95.4	95.9	55.1	95.1
Front_bumper_lower_grille	55.0	52.9	50.6	—	21.7	50.6	54.8	52.4	52.0	51.6	22.3	52.0
Head_lamp(right)	50.8	50.8	62.7	—	34.3	64.2	50.4	50.8	58.6	50.9	34.3	64.2
Head_lamp(left)	47.4	46.2	55.9	—	34.7	65.7	47.5	46.7	55.2	49.5	34.7	67.7
Front_window_glass	94.5	94.6	94.1	—	52.6	94.1	94.5	94.6	93.7	94.1	52.5	93.7
Front_door_glass(right)	45.2	50.5	39.1	—	19.7	62.7	45.5	51.0	39.3	42.8	20.0	63.0
Front_door_glass(left)	31.8	27.9	27.7	—	12.9	51.9	31.8	27.8	27.8	21.5	13.0	51.9

(continued on next page)

Table 9 (continued).

Category	$AP_b@50$ (%)						$AP_s@50$ (%)					
	Mask R-CNN	Cascade Mask R-CNN	MS R-CNN	SOLO-v2	Query-Inst	Our model	Mask R-CNN	Cascade Mask R-CNN	MS R-CNN	SOLO-v2	Query-Inst	Our model
Car_right_door	57.4	63.3	55.2	–	33.3	87.0	57.3	63.3	55.2	53.2	33.4	86.0
Car_left_door	49.8	47.1	46.3	–	30.6	81.4	49.8	47.0	45.7	56.3	30.5	81.1
Front_door_trim(right)	29.2	25.8	30.6	–	9.5	35.9	27.9	16.5	27.3	20.0	10.3	32.9
Front_door_trim(left)	15.3	15.8	14.2	–	8.6	28.3	13.7	10.8	13.6	17.2	9.1	28.1
Front_door_handle(right)	55.1	57.3	42.2	–	23.4	58.7	55.0	56.8	42.2	45.3	24.3	58.1
Front_door_handle(left)	46.8	37.4	29.7	–	14.1	59.0	46.6	37.4	30.9	29.4	14.9	59.4
Fog_lamp(right)	41.4	40.5	35.2	–	20.8	55.5	41.3	40.9	36.6	46.3	21.4	58.6
Fog_lamp(left)	40.5	35.9	41.5	–	19.1	58.5	40.3	36.1	40.7	37.0	19.2	60.0
Front_fender(right)	58.2	67.6	59.5	–	30.1	89.5	58.1	67.6	55.6	46.3	30.1	89.3
Front_fender(left)	52.8	64.8	56.0	–	25.6	83.0	52.8	64.9	56.5	45.1	25.6	82.8
Front_fender_wheel_eyebrow(right)	33.3	28.5	35.7	–	16.3	50.7	31.9	23.1	33.4	30.9	17.0	43.7
Front_fender_wheel_eyebrow(left)	23.3	27.5	30.7	–	11.0	55.4	22.3	23.9	22.7	29.8	11.0	39.7
Exterior_tail_light(right)	52.2	45.6	73.9	–	38.6	79.7	52.2	45.6	67.4	44.3	38.6	79.0
Exterior_tail_light(left)	55.5	77.7	85.0	–	28.8	74.4	55.5	77.7	81.3	50.5	28.9	77.5
Tail_light(right)	47.4	42.1	53.7	–	29.5	71.8	47.3	42.3	49.9	42.5	29.6	71.6
Tail_light(left)	49.3	46.2	50.5	–	32.0	68.1	48.9	46.2	52.1	48.1	32.0	69.2
Fuel_tank_cap	79.3	79.6	80.7	–	49.8	80.7	80.0	79.6	80.6	78.2	50.5	80.6
Grille	96.8	97.0	96.9	–	58.0	96.9	96.8	96.4	95.9	96.7	58.1	95.9
Grille_logo	67.7	67.5	66.9	–	42.4	66.9	67.7	67.5	66.5	67.5	43.2	66.5
Engine_cover	98.3	98.0	98.0	–	61.3	98.0	98.2	98.0	97.4	98.2	61.3	97.4
License_plate	67.6	67.4	65.9	–	41.1	65.9	67.6	67.4	65.8	67.0	41.1	65.8

Note: The bold font represents the optimal value within each row for $AP_b@50$ (%) or $AP_s@50$ (%).

Table 8 demonstrates that our proposed model achieves significantly better performance in $mAP_b@50$ (%) and $mAP_s@50$ (%) compared to other advanced instance segmentation models. $mAP_b@50 : 95$ (%) and $mAP_s@50 : 95$ (%) were used to evaluate the average precision of the bounding boxes and segmentation masks at various IOU thresholds. Specifically, the IOU thresholds ranged from 0.5 to 0.95 with a step size 0.05. The average accuracy of the target bounding boxes and masks was calculated for each IOU threshold. It is worth mentioning that these evaluations were performed under the condition that the confidence score threshold was set to 0.05, consistent with the parameters used during the training phase. The proposed model outperformed other state-of-the-art models in terms of $mAP_b@50 : 95$ (%) and $mAP_s@50 : 95$ (%) metrics. Table 9 displays the category and object box prediction accuracy $AP_b@50$ (%) and instance segmentation accuracy $AP_s@50$ (%) of different vehicle part categories. Our model has a remarkable improvement compared to the baseline and other advanced models for parts with left and right categories.

4.5.4. Orientation scene visualization experiment

To verify the effectiveness of orientation scene classification, this study designed Grad-CAM (Selvaraju et al., 2017) heatmap visualization experiments for orientation scenes. Firstly, the feature map output from the feature extraction network's last layer (layer 4) was obtained. Then, the gradient of the object category for the feature map of the last convolutional layer was calculated. For each channel of the feature map, the global average value of its gradient was calculated to obtain the weight of each channel. The weight of each channel was multiplied by the corresponding feature map to get the weighted feature map. Spatially weighted averaging was performed on the weighted feature maps to obtain the Grad-CAM heatmap for the object category. The Grad-CAM heatmaps for different scene categories for the same image are shown in Fig. 12.

The heatmap for the object category of *Scene1* in Fig. 8 is more informative than the other heatmaps. The predicted important regions are extensive and centralized, indicating that the object category for *Scene1* has a higher proportion of importance, and the classification model is inclined to predict the image as *Scene1*. The final prediction result is *Scene1*, indicating that the orientation scene classification

model performs well. At the same time, there are certain vital regions for the adjacent *Scene2* and *Scene8* of *Scene1*, which indicates that the model may identify them as *Scene2* or *Scene8*, consistent with the recognition patterns in reality.

4.5.5. Vehicle part visualization experiment

To qualitatively analyze the detection performance of our proposed model, Fig. 13 shows the analysis comparison between our proposed model and the baseline model. In Fig. 13(a), the baseline model not only misclassified left and right categories of parts such as Front_fender(left), Mirror_cover(left), Back_door_shell(left), and Head_lamp(left) but also falsely detected the Rear_window_glass part as Front_window_glass. In Fig. 13(b), due to the high confidence threshold, the baseline model had multiple missed detections and also had problems of misclassifying left and right categories of parts such as Front_fender(left) and Head_lamp(left). By introducing the orientation scene and scene understanding module in our proposed model detection results shown in Fig. 13(c), the misclassifications of parts according to left and right categories was effectively addressed. Those that were misdetected in the baseline model such as Front_fender(left), Mirror_cover(left), Back_door_shell(left), and Head_lamp(left) were correctly detected as Front_fender(right), Mirror_cover(right), Back_door_shell(right), and Head_lamp(right), respectively. The multi-detection problem of the same part was also alleviated to some extent by the hierarchical filtering module. With the introduction of the confidence regulation module in our proposed model detection results shown in Fig. 13(d), the problem of missed detections encountered in the baseline model under high confidence threshold conditions was effectively addressed. Meanwhile, the scene understanding module's application resolved the misclassifications of parts according to the left and right categories. In Fig. 13(e), the baseline model mistakenly classified left and right categories of parts such as Exterior_tail_light(left), Rear_fender(left), and Fog_lamp(left). In Fig. 13(g), our proposed model effectively addressed the problem of misclassifications of left and right categories of parts. In Fig. 13(f), there were missed part detections such as Rear_fender(right), Mirror_cover(right), and Fog_lamp(right), as well as some misclassifications of parts according to left and right categories. In Fig. 13(h), our proposed model detection results effectively

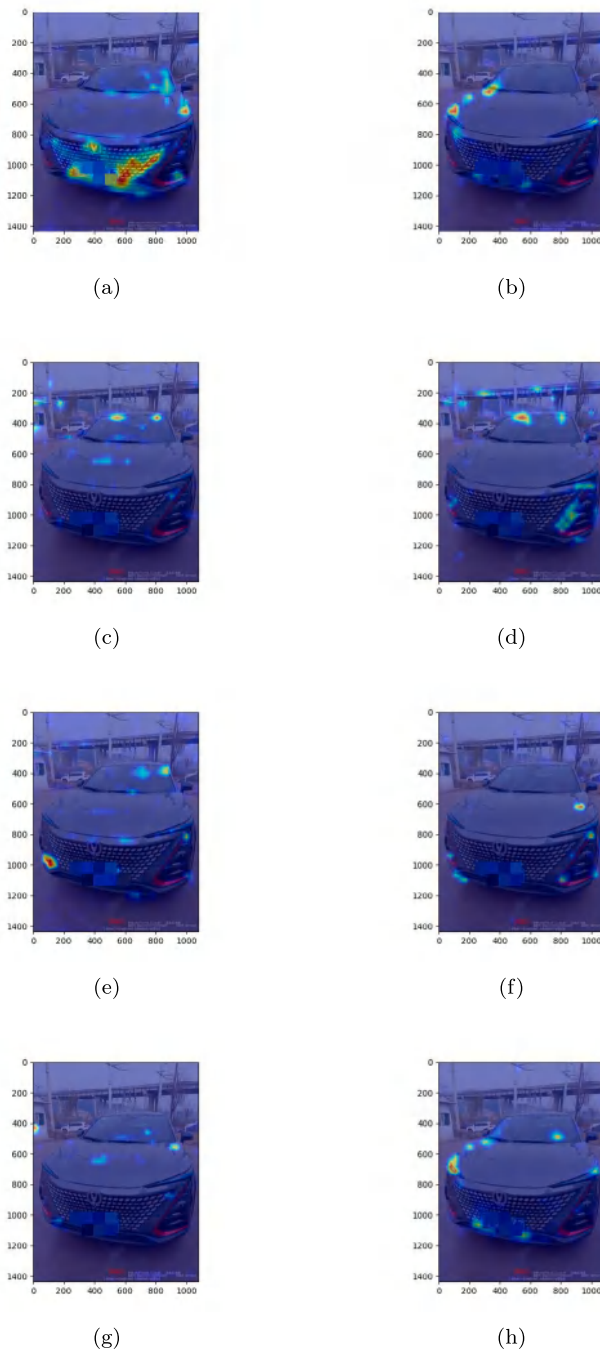


Fig. 12. The same picture was used to analyze the output thermal map results of different scene categories: (a)–(d) The object categories are Scenarios 1–4, (e)–(h) The object category is Scenario 5–8.

addressed missed detections and the problem of misclassifications of some parts according to left and right categories.

5. Conclusion

Aiming at the issues of misclassifications and missed detections caused by left–right and front–rear category recognition errors in vehicle part detection, this paper proposes a scene understanding method

for vehicle part detection, SU-VPDN. Firstly, we introduce the concept of eight-orientation vehicle scenes and use the VOSC network to detect the orientation scene of the vehicle part image. We train a vehicle part instance segment model using VPIS. In SU, during image inference, the part logic correction module corrects and adjusts the low-confidence orientation scene to obtain a high-quality and accurate orientation scene. Then, the part replacement, confidence regulation, and hierarchical filtering modules are used to drive the combination of high-quality and precise orientation scenes and the part baseline detection model to complete the detection task of vehicle parts. These modules effectively address the problems of left–right and front–rear category recognition errors, misclassifications, and missed detections, significantly improving the detection accuracy of the model. The confidence modulation module resolves the issue of missed detections of vehicle parts under high-confidence threshold conditions. Experimental results demonstrate that our proposed model achieves good detection under low-confidence and high-confidence threshold conditions. It effectively addresses the challenges of left/right and front/back category recognition errors, misclassifications, and missed detections in multi-part vehicle detection caused by feature similarities. It outperforms other advanced models, laying a solid foundation for the future fusion of vehicle parts and damage detection. However, it is imperative to acknowledge the inherent limitations of our model. For instance, our approach may need help detecting a close-up image of a vehicle part, making it challenging to discern the corresponding orientation scene, which may result in misclassifications. Future investigations should focus on enhancing the performance of image classification and instance segmentation algorithms alongside the integration of advanced optimization algorithms such as hybrid heuristics (Singh et al., 2022), meta-heuristics (Pasha et al., 2022), and adaptive algorithms (Chen and SF-FWA, 2023; Dulebenets, 2021). These advanced optimization algorithms have proven effective in various domains, including on-line learning, scheduling, multi-objective optimization, transportation, medicine, and data classification. Integrating these algorithms into the SU-VPDN approach can further optimize the performance of the models, allowing for a broader range of applications and contributing to the advancement of intelligent decision systems.

CRedit authorship contribution statement

Yongjie Zhai: Conceptualization, Supervision, Project administration. **Nianhao Chen:** Methodology, Software, Writing – original draft. **Zhenqi Zhang:** Validation, Visualization. **Xunqi Zhou:** Investigation, Writing – review & editing. **Xuan Liu:** Investigation, Writing – review & editing. **Xinying Wang:** Resources. **Qianming Wang:** Data curation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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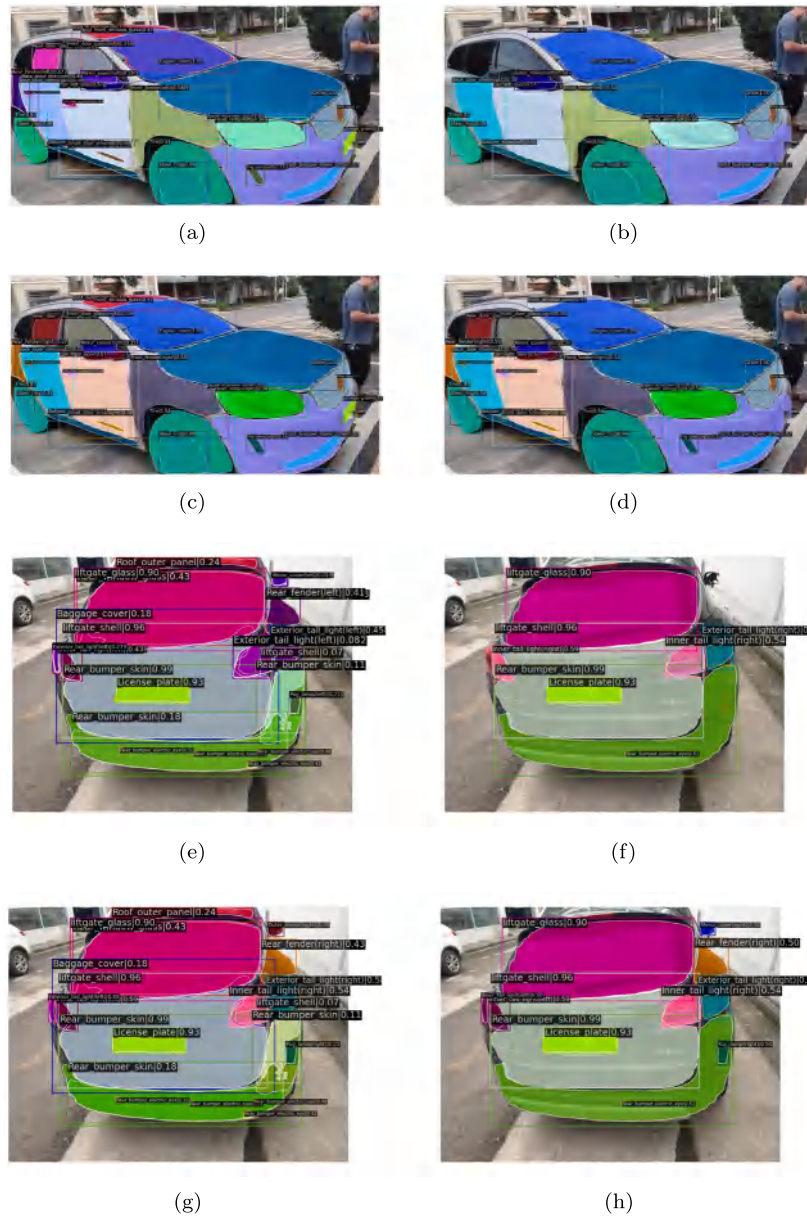


Fig. 13. Comparison of qualitative results of vehicle parts detection: (a)(e) indicates the detection result of baseline model confidence score 0.05, (b)(f) indicates the detection result of baseline model confidence score 0.5, (c)(g) indicates the detection result of Baseline model confidence score 0.05, and (d)(h) indicates the detection result of Baseline model confidence score 0.5.

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